

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 January 2022 Worked Solutions

January 2022 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 January 2022

January 2022 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

$$y = 2^{5-\sqrt{x}}$$

The values of y are given to 3 decimal places.

x	5	5.5	6	6.5	7
y	6.792	6.298	5.858	5.466	5.113

Using the trapezium rule with all the values of y in the given table,

- (a) obtain an estimate for

$$\int_5^7 2^{5-\sqrt{x}} dx$$

giving your answer to 2 decimal places.

(3)

- (b) Using your answer to part (a) and making your method clear, estimate

(i) $\int_5^7 2^{6-\sqrt{x}} dx$

(ii) $\int_5^7 (3 + 2^{5-\sqrt{x}}) dx$

(4)

(Total 7 marks)

Worked Solution - Question 1

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: 17

Final answer

17

Question 2

Applications of Differentiation

Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = 27x^{\frac{1}{2}} - x^{\frac{3}{2}} - 20 \quad x > 0$$

- (a) Find $\frac{dy}{dx}$, giving each term in simplest form.

(2)

- (b) Hence find the coordinates of the stationary point of C .

(4)

- (c) Find $\frac{d^2y}{dx^2}$ and hence determine the nature of the stationary point of C .

(2)

(Total 8 marks)

Worked Solution - Question 2

1. Differentiate and set the condition

Differentiate the expression, then use the gradient, stationary-point, tangent, normal, or optimisation condition.

2. Classify or evaluate

Substitute the required value and use the derivative information to complete the coordinate, gradient, interval, or maximum/minimum.

3. State the final result

Write the answer in the form requested by the question: $-2 \frac{d}{dx} y < 0$ for $x > 0$ / when $x = 9$ so maximum

Final answer

$-2 \frac{d}{dx} y < 0$ for $x > 0$ / when $x = 9$ so maximum

Question 3

Binomial Expansion

3. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{kx}{4}\right)^8$$

where k is a non-zero constant. Give each term in simplest form.

(4)

$$f(x) = (5 - 3x)\left(2 - \frac{kx}{4}\right)^8$$

In the expansion of $f(x)$, the constant term is 3 times the coefficient of x .

- (b) Find the value of k .

(3)

(Total 7 marks)

Leave
blank

Worked Solution - Question 3

1. Choose the required binomial terms

Write the expansion term-by-term and keep only the powers needed for the coefficient or approximation.

2. Compare coefficients

Collect like powers, compare with the given expression, and solve for the required constant.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 4

Laws of Logarithms

$$\log_3(32 - 12x) = 2\log_3(1 - x) + 3$$

(5)

(Total 5 marks)

Worked Solution - Question 4

1. Combine the logarithms

Use the power, product, and quotient laws to rewrite the equation as a single logarithmic or exponential statement.

2. Solve with domain checks

Solve the resulting equation and reject any value that does not satisfy the original logarithms.

3. State the final result

Write the answer in the form requested by the question: carded if seen

Final answer

carded if seen

Question 5

Polynomials

$$f(x) = 3x^2 + 2Ax + 2B + 10$$

where A and B are integers.

Given that

- when $f(x)$ is divided by $(x - 1)$ the remainder is k
- when $f(x)$ is divided by $(x + 1)$ the remainder is $-10k$
- k is a constant

(a) show that

$$11A + 9B = 83$$

(3)

Given also that $(3x - 2)$ is a factor of $f(x)$,

(b) find the value of A and the value of B .

(3)

(c) Hence find the quadratic expression $g(x)$ such that

$$f(x) = (3x - 2)g(x)$$

(2)

(Total 8 marks)

Worked Solution - Question 5

1. Use the polynomial condition

Apply the factor theorem, remainder theorem, or substitution stated in the question to form the required equation.

2. Finish the algebra

Solve the resulting equation and substitute back where needed, then write the requested factor, coefficient, or value.

3. State the final result

Write the answer in the form requested by the question: - +

Final answer

- +

Question 6

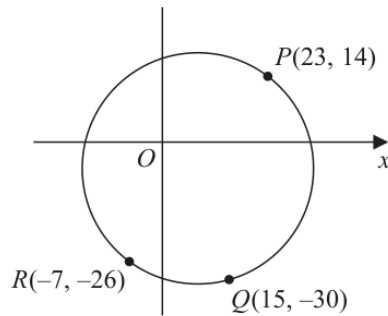


Figure 1

The points $P(23, 14)$, $Q(15, -30)$ and $R(-7, -26)$ lie on the circle C , as shown in Figure 1.

- (a) Show that angle $PQR = 90^\circ$ (2)
- (b) Hence, or otherwise, find
- (i) the centre of C ,
 - (ii) the radius of C . (3)

Given that the point S lies on C such that the distance QS is greatest,

- (c) find an equation of the tangent to C at S , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (3)

(Total 8 marks)

Worked Solution - Question 6

Topic group

1. Identify the circle information

Use the centre-radius form, distance formula, gradient condition, or tangent property required by the diagram.

2. Complete the geometry

Substitute the coordinates carefully and simplify to the requested equation, length, point, or proof.

3. State the final result

Write the answer in the form requested by the question: therefore = - + () 7 18 1 24 y x therefore - = -

Final answer

therefore = - + () 7 18 1 24 y x therefore - = -

Question 7

Trigonometric Equations

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-90^\circ < x < 90^\circ$, the equation

$$3 \sin(2x - 15^\circ) = \cos(2x - 15^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) Solve, for $0 < \theta < 2\pi$, the equation

$$4 \sin^2 \theta + 8 \cos \theta = 3$$

giving your answers to 3 significant figures.

(4)

(Total 8 marks)

Worked Solution - Question 7

1. Rearrange the trigonometric equation

Use the identity or ratio needed to reduce the equation to one trigonometric function in the required interval.

2. List the valid solutions

Find all angles in the given range and discard any extra values outside the interval.

3. State the final result

Write the answer in the form requested by the question: The required value or expression is obtained by the final simplification.

Final answer

The required value or expression is obtained by the final simplification.

Question 8

Geometric Sequences

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blank

8. A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100 mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

- (a) show that the post is driven an **additional** 62 mm into the ground with the 20th hit. (1)
- (b) Find the **total distance** that the post has been driven into the ground after 20 hits. (2)

Given that for each subsequent hit after the 20th hit

- the **additional** distances the post travels form a geometric sequence with common ratio r
- on the 22nd hit, the post is driven an **additional** 60 mm into the ground

- (c) find the value of r , giving your answer to 3 decimal places. (2)

After a total of N hits, the post will have been driven more than 3 m into the ground.

- (d) Find, showing all steps in your working, the smallest possible value of N . (4)

(Total 9 marks)

Worked Solution - Question 8

1. Use the geometric formula

Identify the first term, common ratio, and whether the question needs a term, finite sum, or sum to infinity.

2. Substitute and solve

Apply the correct geometric formula and simplify, checking any validity condition on the ratio.

3. State the final result

Write the answer in the form requested by the question: The required result is shown.

Final answer

The required result is shown.

Question 9

Integration

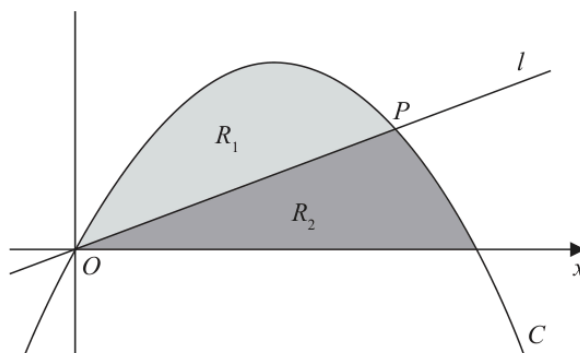


Figure 2

Figure 2 shows

- the curve C with equation $y = x - x^2$
- the line l with equation $y = mx$, where m is a constant and $0 < m < 1$

The line and the curve intersect at the origin O and at the point P .

- (a) Find, in terms of m , the coordinates of P .

(2)

The region R_1 , shown shaded in Figure 2, is bounded by C and l .

- (b) Show that the area of R_1 is

$$\frac{(1-m)^3}{6}$$

(5)

The region R_2 , also shown shaded in Figure 2, is bounded by C , the x -axis and l .

Given that the area of R_1 is equal to the area of R_2

- (c) find the exact value of m .

(3)

(Total 10 marks)

Worked Solution - Question 9

1. Set up the integral

Choose the correct integrand and limits, or apply the trapezium rule if the question gives tabulated values.

2. Evaluate the area or expression

Integrate or calculate the trapezium estimate carefully, then simplify the requested value.

3. State the final result

Write the answer in the form requested by the question: $= + + - 3 \ 1 \ 1 \ 2 \text{ m} = -$

Final answer

$= + + - 3 \ 1 \ 1 \ 2 \text{ m} = -$

Question 10

Proof

“if p is a prime number then $2p + 1$ is also a prime number”

is not true.

(1)

(ii) Use proof by exhaustion to prove that if n is an integer then

$$5n^2 + n + 12$$

is always even.

(4)

Q10

(Total 5 marks)

(Total 5 marks)

Worked Solution - Question 10

Topic group

1. Set up the proof

Translate the statement into algebra, or choose a clear counterexample if the statement is false. Keep each implication justified.

2. Complete the argument

Simplify the algebra or evaluate the counterexample, then finish with a clear conclusion.

3. State the final result

Write the answer in the form requested by the question: even and 2 even : $20 \leq 12n + k + k +$ which is even

Final answer

even and 2 even : $20 \leq 12n + k + k +$ which is even